

Production Process Specification

VOLTRA MOBILITY // MANUFACTURING PROCESS SPECIFICATION

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Classification: Internal Manufacturing Standard

1. EXECUTIVE OVERVIEW

Voltra Mobility operates a continuous synchronous automotive production line for two

- Voltra Model V-1 [All-electric urban passenger sedan, target 500 units/day]
- Voltra Commercial Fleet Van [Light commercial delivery van, target 150 units/day]

2. ASSEMBLY LINE ARCHITECTURE & DEPENDENCIES

Line 1: Powertrain Integration

- Integrates Electric Traction Motor Stator assemblies and high-voltage battery modul
- Output: Complete high-voltage chassis powertrain unit. Feeds directly into Line 3.

Line 2: ECU Sub-assembly & Avionics

- Primary Machine: SMT Robot #4 [Surface Mount Technology high-speed placer].
- Required Component: Microcontroller MCU-900 [Supplied exclusively by Apex Microelec
- Machine B [SMT Robot #4] places MCU-900 chipsets onto multilayer motherboard PCBs.
- Output: Vehicle Central Avionics and Electronic Control Unit [ECU Sub-assembly].
- Dependency: Line 2 output feeds directly into Final Vehicle Synthesis on Line 3.

Line 3: Final Vehicle Synthesis

- Merges chassis, Line 1 powertrain assemblies, Line 2 ECU modules, and sensor harness
- Outputs finished commercial products: Voltra Model V-1 and Voltra Commercial Fleet
- CRITICAL PATH: Failure of Line 2 ECU Sub-assembly halts Line 3 final road-readiness